# Pranav Waghanna

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## Education

**University of Rochester**, Master of Science in Computer Science — Aug 2024 – Dec 2025

**Pune Institute of Computer Technology**, Bachelor in Information Technology — Aug 2020 – Jun 2024

## Projects

**eBPF System and File Access Monitor** (View Project) — Nov 2025 – Jan 2026

- Developed a security tool on Linux that watches how programs access files and the network, using C, extended Berkeley Packet Filter(eBPF), Python and TCP/IP stack concepts to detect unusual activity in real time.
- Leveraged LOSS development tools (gcc, strace, valgrind) to evaluate syscall tracing overhead, identifying a 2× latency increase while maintaining minimal CPU and memory usage for efficient performance tuning.
- Used Linux internals with eBPF (BCC) in a Bodhi Linux virtual machine to trace syscalls, capturing 14K+ events in 32 seconds (447 events/sec) with zero loss for high-performance monitoring.
- Evaluated runtime overhead using syscall tracing techniques and identifying a 2x latency increase in 'ls' execution (0.015s to 0.031s) while maintaining a minimal CPU and memory footprint.

**Houdini: VFS-layer Rootkit for FreeBSD** (View Project) — Sept 2025 – Dec 2025

- Built a stealth system program using C/C++ on a Unix/Linux-like system (FreeBSD) that hides files and spoofs system data, leveraging kernel-level development concepts to evade standard system visibility.
- Used Unix internals and kernel-level development to hook VFS-layer functions (e.g., getdirents64()), concealing 6 targeted files and spoofing directory contents returned to userspace tools for stealth persistence.
- Applied Linux POSIX mechanisms (IPC, threading) to manipulate process and file visibility while maintaining consistent fake outputs, ensuring normal system behavior despite underlying tampering.
- Utilized LOSS development tools (gcc, strace) to build and validate kernel interactions, bypassing syscall-based monitoring by modifying VFS function pointers for a more stealthy compromise vector.

**Roomsense: AR Classroom Engagement System** (Website) — Sept 2025 – Dec 2025

- Built a full-stack AR system integrating Snap Spectacles, a web platform, MongoDB, and Flask to deliver real-time engagement visualization at 20 FPS with sub-1s UI update latency.
- Led the development of on-device engagement estimation and integrated facial landmark analysis across 10+ signals per frame with a team of 4 while preserving user privacy.
- Orchestrated user testing with eight participants to evaluate the software system that led to a perfect 100% task completion rate and an average satisfaction score of 8.75/10, uncovering previously unknown needs.

## Experience

**Backend Developer** - Freelance Contract, Brydge — Feb 2026 – Present

- Built a user-friendly web application using HTML, CSS, JavaScript, and ReactJS to help users easily discover and book social experiences, improving engagement and achieving fast mobile load times under 2 seconds.
- Developed scalable backend services using Node.js and Express with PostgreSQL(Supabase) and Cloudflare R2's S3 containers, and implemented a CI/CD pipeline with GitHub to automate deployments, reducing manual admin work from hours to minutes and ensuring faster releases.
- Designed and integrated REST APIs for event management and booking workflows, enabling secure data exchange between frontend and backend and improving system efficiency and reliability.
- Leveraged AI/LLM tools to optimize development workflows and enhance SEO strategies, increasing platform discoverability and contributing to Rs. 2L+ revenue within the first 4 weeks.
- Engineered a secure booking system with validation and QR-based UPI payments, along with automated email notifications, improving booking completion rates by 20

## Skills

**Languages:** C/C++, Python, Java, Rust, SQL, JavaScript, HTML, CSS, Shell, Bash.
**Technologies:** POSIX, pthreads, mmap, perf, Valgrind, GDB, eBPF, Kernel Programming, Node.js, Express, PostgreSQL, REST APIs, CI/CD, AWS (EC2, S3), TCP/IP, DNS, Sockets, Linux/Unix